

Sec: NAITS
Time: 3 Hrs.

CTM-01

Date: 22-02-2024
Max. Marks: 300

PHYSICS

Section	Question Type	+Ve Marks	-Ve Marks	No. of Qs.	Total marks
Sec - I (Q.N : 1 - 20)	Questions with Single Correct Choice	4	-1	20	80
Sec-II (Q.N : 21 - 30)	Questions with Numerical Answer Type (+ / - Number)	4	-1	10	20
Total				30	100

CHEMISTRY

Section	Question Type	+Ve Marks	-Ve Marks	No. of Qs.	Total marks
Sec - I (Q.N : 31 - 50)	Questions with Single Correct Choice	4	-1	20	80
Sec-II (Q.N : 51 - 60)	Questions with Numerical Answer Type (+ / - Number)	4	-1	10	20
Total				30	100

MATHEMATICS

Section	Question Type	+Ve Marks	-Ve Marks	No. of Qs.	Total marks
Sec-I (Q.N : 61 - 80)	Questions with Single Correct Choice	4	-1	20	80
Sec-II (Q.N : 81 - 90)	Questions with Numerical Answer Type (+ / - Number)	4	-1	10	20
Total				30	100

Exam Syllabus

PHYSICS	COMPLETE XI SYLLABUS
CHEMISTRY	COMPLETE XI SYLLABUS
MATHEMATICS	COMPLETE XI SYLLABUS

PHYSICS

Max. Marks: 100

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

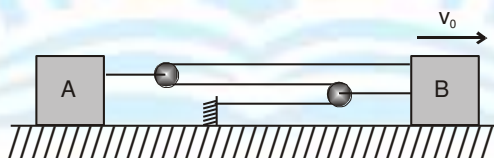
1. A particle at a height 'h' from the ground is projected with an angle 30° from the horizontal, it strikes the ground making angle 45° with horizontal. It is again projected from the same point at height h with the same speed but with an angle of 60° with horizontal. Find the angle it makes with the horizontal when it strikes the ground :

A) $\tan^{-1}(4)$ B) $\tan^{-1}(5)$ C) $\tan^{-1}(\sqrt{5})$ D) $\tan^{-1}(\sqrt{3})$

2. A man crosses the river perpendicular to river flow in time t seconds and travels an equal distance down the stream in T seconds. The ratio of man's speed in still water to the speed of river water will be :

A) $\frac{t^2 - T^2}{t^2 + T^2}$ B) $\frac{T^2 - t^2}{T^2 + t^2}$ C) $\frac{t^2 + T^2}{t^2 - T^2}$ D) $\frac{T^2 + t^2}{T^2 - t^2}$

3. Block B moves to the right with a constant velocity v_0 . The velocity of block A relative to B is :



A) $\frac{v_0}{2}$, towards left B) $\frac{v_0}{2}$, towards right
C) $\frac{3v_0}{2}$, towards left D) $\frac{3v_0}{2}$, towards right

4. A block of mass 50 kg is projected horizontally on a rough horizontal floor. The coefficient of friction between the block and the floor is 0.1. The block strikes a light spring of stiffness $k = 100$ N/m with a velocity 2m/s. The maximum compression of the spring is :

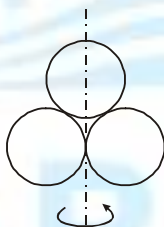
A) 1 m B) 2 m C) 3 m D) 4 m

5. The horizontal and vertical distance travelled by a particle in time t are given by $x = 6t$ and $y = 8t - 5t^2$. $g = 10 \text{ m/sec}^2$, then the initial speed of the particle is:-

A) 8 m/sec B) 10 m/sec C) 5 m/sec D) Zero

Rough Work

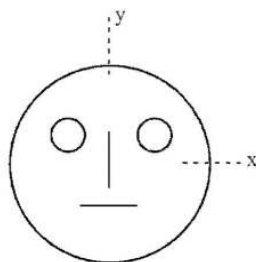
6. A car starts from rest travelling with constant acceleration. If distance covered by it in 10th second of its journey is 19 m, what will be the acceleration of car:-
 A) 4 m/s^2 B) 3 m/s^2 C) 2 m/s^2 D) 1 m/s^2
7. A 20 kg block is resting on a horizontal floor. A force of 75 N is required to just move the block and a force of 60 N is required to move the block with constant velocity then coefficient of static friction is :-
 A) 0.38 B) 0.44 C) 0.52 D) 0.60
8. A 5 kg block is sliding with constant velocity on an inclined plane of inclination 30° then coefficient of friction is:-
 A) $\frac{1}{\sqrt{3}}$ B) $\frac{2}{\sqrt{3}}$ C) $\sqrt{3}$ D) $2\sqrt{3}$
9. The radius to which the present radius of earth be compressed so that the escape velocity is increased 10 times is –
 A) 6.4 km B) 64 km C) 640 km D) 4800 km
10. Three rings each of mass m and radius r are so placed that they touch each other. The radius of gyration of the system about the axis as shown in the figure is :



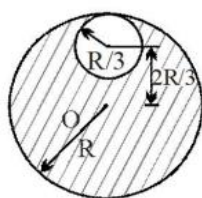
- A) $\sqrt{\frac{6}{5}} r$ B) $\sqrt{\frac{5}{6}} r$ C) $\sqrt{\frac{6}{7}} r$ D) $\sqrt{\frac{7}{6}} r$

Rough Work

11. Look at the drawing given in the figure which has been drawn with ink of uniform line-thickness. The mass of ink used to draw each of the two inner circles, and each of the two line segments is m . The mass of the ink used to draw the outer circle is $6m$. The coordinates of the centres of the different parts are: outer circle $(0,0)$, left inner circle $(-a,a)$, right inner circle (a, a) , vertical line $(0, 0)$ and horizontal line $(0, -a)$. The y-coordinate of the center of mass of the ink in this drawing is



- A) $\frac{a}{10}$ B) $\frac{a}{8}$ C) $\frac{a}{12}$ D) $\frac{a}{3}$
12. A disc has mass $9m$. A hole of radius $R/3$ is cut from it as shown in the figure. the moment of inertia of remaining part about an axis passing through the centre 'O' of the disc and perpendicular to the plane of the disc is:

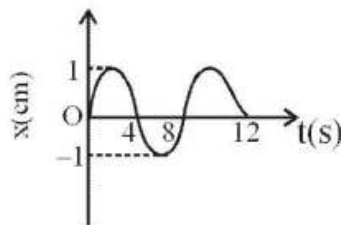


- A) $8mR^2$ B) $4mR^2$ C) $\frac{40}{9}mR^2$ D) $\frac{37}{9}mR^2$
13. If two tuning forks A and B are sounded together, they produce 4 beats per second. A is then slightly loaded with wax, they produce less beats when sounded again. The frequency of A is 256 Hz. The frequency of B will be :-
- A) 250 Hz B) 252 Hz C) 260 Hz D) 262 Hz

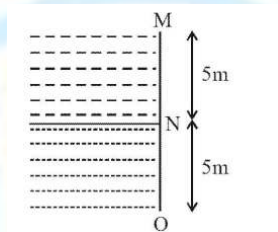
Rough Work

14. The x-t graph of a particle undergoing SHM is shown below. The acceleration of the particle at $t = 8/3$ s is :-

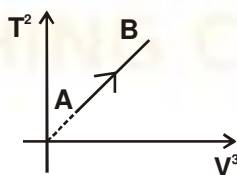
- A) $\frac{\sqrt{3}}{32} \pi^2 \text{ cm/s}^2$ B) $-\frac{\pi^2}{32}$
C) $\frac{\pi^2}{32} \text{ cm/s}^2$ D) $-\frac{\sqrt{3}}{32} \pi^2 \text{ cm/s}^2$



15. Liquids of densities ρ_1 and ρ_2 ($\rho_2 = 2\rho_1$) are filled up behind a square wall of side 10 m as shown in figure. Each liquid has a height of 5 m. The ratio of the forces due to these liquids exerted on upper part MN to that at the lower part NO is (Assume that the liquids are not mixing)



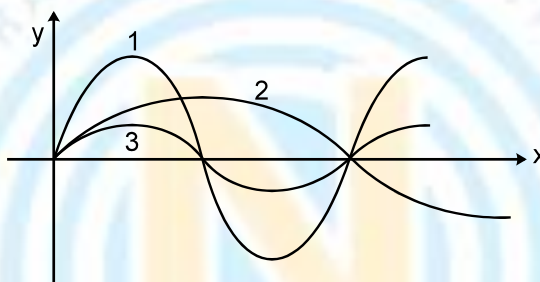
- A) 1/4 B) 2/3 C) 1/3 D) 1/2
16. To what depth below the surface of sea should a rubber ball be taken as to decrease its volume by 0.1% ? [Take : density of sea water = 1000 kgm^{-3} , Bulk modulus of rubber = $9 \times 10^8 \text{ Nm}^{-2}$; acceleration due to gravity = 10 ms^{-2}]
- A) 9 m B) 18 m C) 180 m D) 90 m
17. If ideal diatomic gas follows the process, as shown in graph, where T is temperature in kelvin and V is volume (m^3), then molar heat capacity for this process will be [in terms of gas constant R] :



- A) $\frac{7R}{2}$ B) 5R C) $\frac{19R}{6}$ D) $\frac{11R}{2}$

Rough Work

18. The molar heat capacity at constant pressure of nitrogen gas at STP is nearly $3.5 R$. Now when the temperature is increased, it gradually increases and approaches $4.5 R$. The most appropriate reason for this behavior is that at high temperatures
- nitrogen does not behave as an ideal gas
 - nitrogen molecules dissociate in atoms
 - the molecules collides more frequently
 - molecular vibration gradually become effective
19. Graph shows three waves that are separately sent along a string that is stretched under a certain tension along x -axis. If ω_1 , ω_2 and ω_3 are their angular frequencies respectively then :



- $\omega_1 = \omega_3 > \omega_2$
 - $\omega_1 > \omega_2 > \omega_3$
 - $\omega_2 > \omega_1 = \omega_3$
 - $\omega_1 = \omega_2 = \omega_3$
20. A capillary tube of radius R is immersed in water and water rises in it to a height H . Mass of water in capillary tube is M . If the radius of the tube is doubled, mass of water that will rise in capillary tube will be

- $2M$
- M
- $\frac{M}{2}$
- $4M$

Rough Work

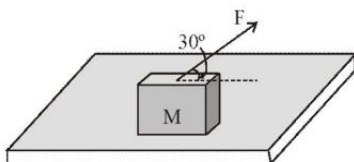
SECTION-II

(Numerical Value Answer Type)

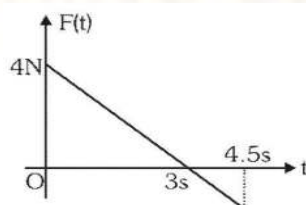
This section contains 10 questions. The answer to each question is a Numerical value. If the Answer is in the decimals, **Mark the nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

21. Two rods of same dimensions, but made of different materials are joined end to end with their free ends being maintained at 100°C and 0°C respectively. The temperature of the junction is 70°C . Then the temperature of the junction if the rods are interchanged will be equal to $T^{\circ}\text{C}$ Find T :
22. The time period of a satellite of earth is 5 hours. If the separation between the earth and the satellite is increased to 4 times the previous value, the new time period becomes (in hrs).
23. A body is started from rest with acceleration 2 m/s^2 till it attains the maximum velocity then retards to rest with 3 m/s^2 . If total time taken is 10 second then maximum speed attained is :-
24. A force of 40 N is acting on a block of mass 5 kg as shown in figure. If coefficient of friction is 0.2 then acceleration of block is :- (convert your answer in nearest integer)
($g = 10\text{ m/sec}^2$)

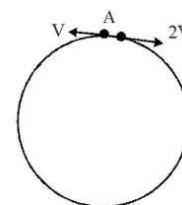


25. A block of mass 2 kg is free to move along the x-axis. IT is at rest and from $t = 0$ onwards it is subjected to a time-dependent force $F(t)$ in the x-direction. The force $F(t)$ varies with t as shown in the figure. The kinetic energy (in J) of the block after 4.5 second is (convert your answer in nearest integer)

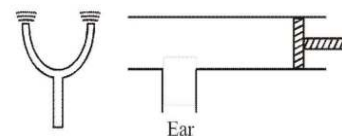


Rough Work

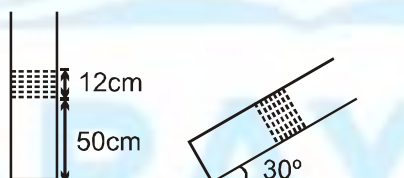
26. Two small particles of equal masses start moving in opposite direction from a point A in a horizontal circular orbit. Their tangential velocities are v and $2v$, respectively, as shown in the figure. Between collisions, the particles move with constant speeds. After making how many elastic collisions, other than that at A, these two particles will again reach the point A?



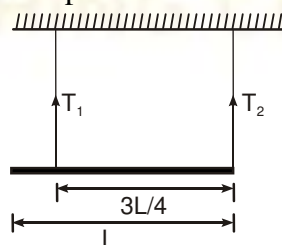
27. A vibrating tuning fork of frequency n is placed near the open end of a long cylindrical tube. The tube has a side opening and is also fitted with a movable reflecting piston. As the piston is moved through 8.75 cm, the intensity of sound changes from a maximum to minimum. If the speed of sound is 350 metre per second, then n is :-



28. A damped harmonic oscillator has a frequency of 5 oscillations per second. The amplitude drops to half its value for every 10 oscillations. The time it will take to drop to $\frac{1}{1000}$ of the original amplitude is close to :-
29. In given figure, an ideal gas is trapped between a mercury column and the closed end of a uniform vertical tube. The upper end of the tube is open to the atmosphere. Initially the lengths of the mercury column and the trapped air column are 12 cm and 50 cm respectively. When the tube is tilted slowly in a vertical plane through an angle of 30° with horizontal then the new length of air column is $\frac{x}{41}$ m. Find x . Assuming the temperature to remain constant. ($P_{\text{atm}} = 76$ cm of Hg)



30. A uniform rod of mass m and length L is suspended with two massless strings as shown in the figure. If the rod is at rest in a horizontal position the ratio of tension in the two strings T_1/T_2 is:



Rough Work

CHEMISTRY

Max. Marks: 100

SECTION – I (SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

31. If 5g, H_2 is mixed with 14 gm nitrogen gas for the following reaction $N_2 + 3H_2 \rightarrow 2NH_3$ at the end mass of H_2 left unreacted is.

A) 2 gm B) 4 gm C) 6 gm D) 8 gm

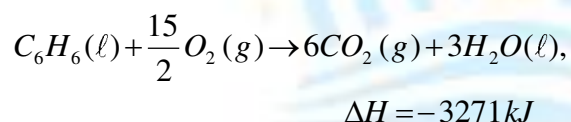
32. The number of waves made by a Bohr electron in an orbit of maximum magnetic quantum number +2

A) 3 B) 4 C) 2 D) 1

33. The oxidation state of chromium in the final product formed by the reaction between KI and acidified dichromate solution is :

A) +3 B) +6 C) +5 D) +4

34. Consider the reaction at 300 K



What is ΔU for the combustion of 1.5 mole of benzene at $27^\circ C$?

A) $-326.25 kJ$ B) $-4900.25 kJ$
C) $-4906.5 kJ$ D) $-3274.75 kJ$

35. For adiabatic free expansion of ideal gas correct option will be :-

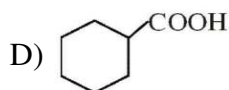
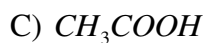
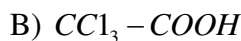
A) $q=0, \Delta T \neq 0, w=0$ B) $q \neq 0, \Delta T=0, w=0$
C) $q=0, \Delta T=0, w=0$ D) $q=0, \Delta T < 0, w < 0$

36. A weak acid of dissociation constant 10^{-5} is being titrated with aqueous NaOH solution. The PH at the point of one-third neutralization of the acid will be:

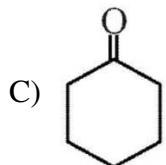
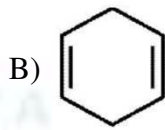
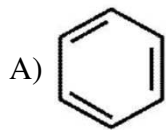
A) $5 + \log 2 - \log 3$ B) $5 - \log 2$
C) $5 - \log 3$ D) $5 - \log 6$

Rough Work

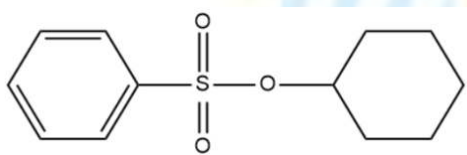
37. Which of the following compound is most acidic in nature?



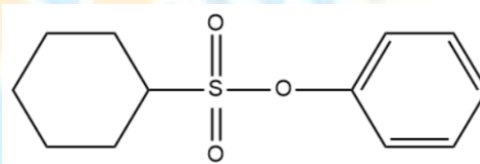
38. Least stable compound is



39. The given compounds show which type of isomerism?



and



A) Chain isomerism

B) Positional isomerism

C) Metamerism

D) Functional group isomerism

40. Which of the following alkanes has the lowest boiling point?

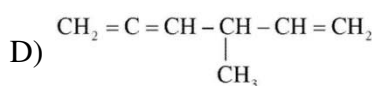
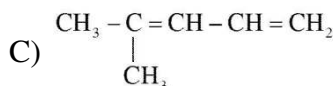
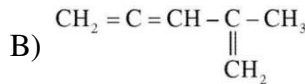
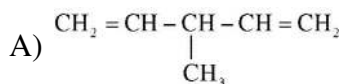
A) n-Pentane

B) Isopentane

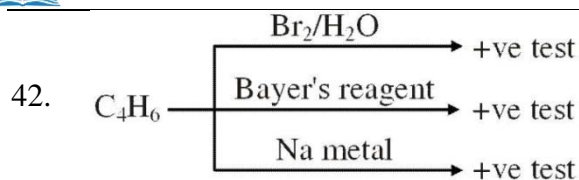
C) Neopentane

D) n-Hexane

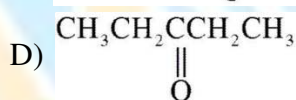
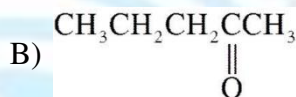
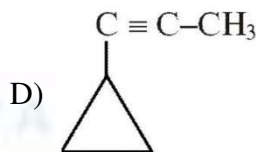
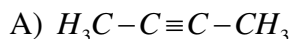
41. An alkene given two moles of HCHO , one mole of CO_2 and one mole of $\text{CH}_3 - \text{C}(=\text{O}) - \text{CHO}$ on ozonolysis. What is its structure?



Rough Work



Compound X is :-



44. Total number of p block elements among given atomic number 83, 79, 42, 64, 37, 54, 34 :-

A) 3

B) 4

C) 5

D) 6

45. General electronic configuration of outermost and penultimate shell is

$(n-1)s^2 (n-1)p^6 (n-1)d^x ns^2$. If $n = 4$ and $x = 5$, then number of protons in the nucleus will be :-

A) > 25

B) < 25

C) 25

D) 30

46. An ion M^{+3} has electronic configuration $[Ar] 3d^{10} 4s^2$ element M belongs to :-

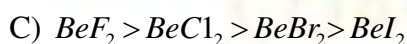
A) s-block

B) p-block

C) d-block

D) f-block

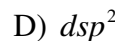
47. Which of the following is not a correct order of solubility in water :-



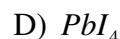
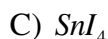
48. Which show highest lattice energy :-



49. The type of hybridisation of boron in diborane is



50. Which of the following halides is least stable and has doubtful existence ?



Rough Work

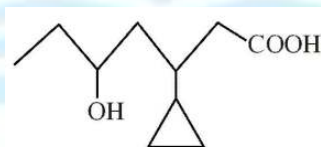
SECTION-II

(Numerical Value Answer Type)

This section contains 10 questions. The answer to each question is a Numerical value. If the Answer is in the decimals, **Mark the nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

51. $A + B \rightarrow C + D$ initially the concentration of A and B both are equal but at the equilibrium concentration of 'D' will be twice of that of A, then what will be the equilibrium constant of the reaction?
52. For the redox reaction
 $x \text{Zn} + y \text{NO}_3^- \rightarrow \text{Zn}^{2+} + \text{NH}_4^+ + z \text{OH}^-$ in basic medium, sum of (x+y+z) coefficients of Zn , NO_3^- , and OH^- in the balanced equation respectively is
53. What is the final temperature (in kelvin) of 0.10 mole monoatomic ideal gas that performs 75 cal of work adiabatically if the initial temperature is 227°C ?
 (use $R = 2 \text{ cal/K-mol}$)
54. A weak acid 0.1M HA is 1% ionised, pH of the solution will be –
55. How many carbon atom in parent carbon-chain in given compound.



56. How many alcohol isomers are possible for given compound $\text{C}_4\text{H}_{10}\text{O}$.
57. The number of double bonds in gammexane will be
58. The number of d electrons in Fe^{+2} are
59. The radius(in $\text{\AA}$) of the second bohr orbit for hydrogen atom is____(Answer in nearest integer)
60. Find out the value of 'X' in ion $[\text{H}_x\text{O}_4]^+$:

Rough Work

MATHEMATICS

Max. Marks: 100

SECTION – I (SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

61. A complex number z is said to be unimodular if $|z| = 1$. Suppose z_1 and z_2 are complex numbers such that $\frac{z_1 - 2z_2}{2 - z_1\bar{z}_2}$ is unimodular and z_2 is not unimodular. Then the point z_1 lies

A) straight line parallel to x-axis
B) straight line parallel to y-axis

C) circle of radius 2
D) circle of radius $\sqrt{2}$

62. In a triangle ABC with fixed base BC, the vertex A moves such that $\cos B + \cos C = 4 \sin^2 \frac{A}{2}$. If a , b and c denote the length of the sides of the triangle opposite to the angles A, B and C respectively, then

A) $b + c = 4a$
B) $b + c = 2a$

C) locus of points A is a hyperbola
D) locus of point A is a pair of straight lines

63. Number of common tangents of the circles $(x+2)^2 + (y-2)^2 = 49$ and $(x-2)^2 + (y+1)^2 = 4$ is:

A) 0
B) 1
C) 2
D) 3

64. If a, b, c are in H.P. then the value of $\frac{b+a}{b-a} + \frac{b+c}{b-c} - 2$ is

A) 2
B) 3
C) 4
D) 0

65. Let $z = x + iy$ be a complex number where x and y are integers. Then the area of the rectangle whose vertices are the roots of the equation $z\bar{z}^3 + \bar{z}z^3 = 350$ is__

A) 48
B) 32
C) 40
D) 80

66. If the distance of the centre of the ellipse $4(x - 2y + 1)^2 + 9(2x + y + 2)^2 = 25$ from the origin is λ times its eccentricity, then $5\lambda^2$ is :

A) 7
B) 6
C) 8
D) 9

Rough Work

67. Let α and β be the roots of equation $px^2 + qx + r = 0$, $p \neq 0$. If p, q, r are in the A.P. and $\frac{1}{\alpha} + \frac{1}{\beta} = 4$, then the value of $|\alpha - \beta|$ is :
- A) $\frac{\sqrt{34}}{9}$ B) $\frac{2\sqrt{13}}{9}$ C) $\frac{\sqrt{61}}{9}$ D) $\frac{2\sqrt{17}}{9}$
68. Let P be the point on the parabola, $y^2 = 8x$ which is at a minimum distance from the centre C of the circle, $x^2 + (y + 6)^2 = 1$. Then the equation of the circle, passing through C and having its centre at P is :
- A) $x^2 + y^2 - x + 4y - 12 = 0$ B) $x^2 + y^2 - \frac{x}{4} + 2y - 24 = 0$
C) $x^2 + y^2 - 4x + 9y + 18 = 0$ D) $x^2 + y^2 - 4x + 8y + 12 = 0$
69. Two vertices of a triangle are $(5, -1)$ and $(-2, 3)$. If orthocentre is the origin then co-ordinates of the third vertex are
- A) $(7, 4)$ B) $(-7, -4)$ C) $(4, 7)$ D) $(-4, -7)$
70. The sum of all the real roots of the equation $(x - 2)^2 + |x - 2| - 2 = 0$ is _____
- A) 3 B) 2 C) -3 D) 4
71. If $x = \sum_{n=0}^{\infty} a^n$, $y = \sum_{n=0}^{\infty} b^n$, $z = \sum_{n=0}^{\infty} c^n$ where a, b, c are in AP and $|a| < 1$, $|b| < 1$, $|c| < 1$, then x, y, z are in :
- A) HP B) Arithmetico-Geometric Progression
C) AP D) GP
72. A hyperbola passes through the point $P(\sqrt{2}, \sqrt{3})$ and has foci at $(\pm 2, 0)$. Then the tangent to this hyperbola at P also passes through the point :
- A) $(3\sqrt{2}, 2\sqrt{3})$ B) $(2\sqrt{2}, 3\sqrt{3})$
C) $(\sqrt{3}, \sqrt{2})$ D) $(-\sqrt{2}, -\sqrt{3})$
73. The focus of the parabola is $(1, 1)$ and the tangent at the vertex has the equation $x + y = 1$. Then:
- A) length of the latus rectum is $2\sqrt{2}$
B) equation of the parabola is $(x - y)^2 = 4(x + y - 1)$
C) the co-ordinates of the vertex are $(1/2, 1/2)$
D) All of these

Rough Work

Rough Work

SECTION-II

(Numerical Value Answer Type)

This section contains 10 questions. The answer to each question is a Numerical value. If the Answer is in the decimals, **Mark the nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

81. The sum of digits in the co-efficient of x in the expansion of $(1-2x^3 + 3x^5) \left(1 + \frac{1}{x}\right)^8$ is__
82. The circle $x^2 + y^2 - 6x - 10y + \lambda = 0$ neither touches nor intersect the coordinate axis and the point $(1,4)$ lies inside the circle then maximum integral value of λ is__
83. If z_1 and z_2 are two non-zero complex numbers such that $|z_1 + z_2| = |z_1| + |z_2|$, then $\arg z_1 - \arg z_2$ is equal to __
84. If $\sin t + \cos t = \frac{1}{5}$ then absolute modulus of integral value of $\tan \frac{t}{2}$ is equal to__
85. The absolute sum of maximum and minimum value of the expression $f(x) = 1 + 2\sin x + \cos^2 x$ is λ , then $[\lambda]$ (where $[x]$ denotes greatest integer less than or equal to x) is__
86. The least possible integral value of x satisfying the inequality $\sqrt{x^2 - x - 6} < 2x - 3$ is__
87. The locus of the vertices of the family of parabolas $y = \frac{a^3 x^2}{3} + \frac{a^2 x}{2} - 2a$ is $xy = k$ where k is a rational number of the type $\frac{p}{q}$ where $\gcd(p, q) = 1$, then $|q - p| =$
88. Number of positive integral solutions of $xyz = 21600$ is λ then number of proper divisors of λ are__
89. Number of ways in which 3 persons throw a normal die to have a total score of 11 is N then number of ways in which $\left(\frac{N}{9} + 2\right)$ letters can be inserted into $\left(\frac{N}{9} + 2\right)$ envelopes so that none of the letter goes to its corresponding envelope__
90. if the letters of the word INDIAN are permuted and arranged in the dictionary then the rank of the word INDIAN is__

Subject	Physics	Chemistry	Mathematics
Setter Branch	SURAT		
SME Name	MR. MIHIR SHAH	MR. SUNNY DUTTA	MR. ANKIT SHARMA
Phone Number	89056 01062	95606 70699	88508 54709

Rough Work